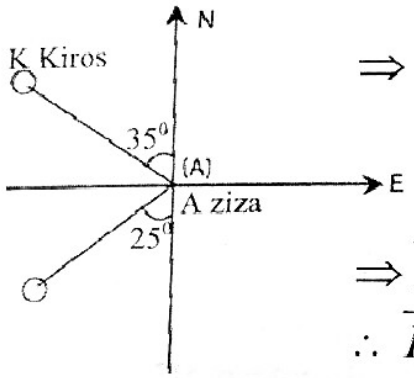
A. 40m B. $10\sqrt{19}m$ C. $10\sqrt{7}m$ D. 25m

Solution: In ΔAKG, the angle formed



$$m(\angle AKG) = 180^0 - (35^0 + 25^0) = 120^0$$

$$\Rightarrow \overline{KG}^2 = \overline{AG}^2 + \overline{AK}^2 - 2(\overline{AG})(\overline{AK})\cos 128$$

$$= 30^2 + 20^2 - 2(30)(20)\left(\frac{-1}{2}\right)$$

$$\Rightarrow \overline{KG} = \sqrt{900 + 400 + 600}$$

$$\therefore \overline{KG} = \sqrt{1900} = 10\sqrt{19},$$

Answer: B

Simple Harmonic Motion

The periodic nature of the trigonometric function is useful for describing the motion of a point on an object that vibrates, oscillates or moved by wave motion.

Examples of periodic quanties are

* Pendulum, spring, periodic fluctuation in the population of specias. Many of these quantities described by sine or cosine function, and is called simple harmonic motion.

A point that moves on a coordinate line is said to be in simple harmonic motion if its distance from the origin at time t is given by either.

$d(t) = a \sin wt$ or $d(t) = a\cot wt$

* amplitude, $|a|$

* period, $T = \frac{2\pi}{w}$

* frequency $f = \frac{1}{T} = \frac{w}{2\pi}$

59. Given the equation for simple harmonic motion

$$d = 6\cos\frac{3\pi t}{4}$$

Find

a) the maximum displacement
b) the frequency
c) the value of d when t = 4
d) the least positive value of t for which d = 0

Solution:

a) maximum displacement, $|a| = 6$

b) frequency, $f = \frac{w}{2\pi} = \frac{\frac{3\pi}{4}}{2\pi} = \frac{3}{8}$

c) $d(4) = 6\cos\left(\frac{3\pi}{4}(4)\right) = 6\cos 3\pi = 6(-1) = -6$

d) To find the least positive value of t for which d = 0, solve the equation $d = 6\cos\frac{3\pi}{4}t = 0$ to obtain

$$\frac{3\pi}{4}t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \Rightarrow t = \frac{2}{3}, 2, \frac{10}{3}$$

So the least positive value of t is $t = \frac{2}{3}$

60. The monthly sales (in thousands of unit) of a seasonal product in (ECX) is modeled by

$S(t) = 56 + 9\sin\left(\frac{3\pi t}{5}\right);\ 0 \le t \le 12,$ t is time in month

Determine:

a) the initial sale
b) the largest and smallest sale
c) the first time in which the sale reaches 51,500 Birr
d) the sale after one year

Solution:

a) The initial sale when t = 0, S(0) = 56
⇒ The initial sale, is 56,000 birr

b) The sale is largest if $\sin\left(\frac{3\pi t}{5}\right) = 1$ and

It is smallest if $\sin\left(\frac{3\pi t}{5}\right) = -1$

⇒ maximum sale = (56 + 9(1))1000 = 65,000 Birr
⇒ minimum sale = (56 + 9(-1))1000 = 47,000 Birr

c) $S(t) \times 1000 = 51{,}500 \Rightarrow 56 + 9\sin\left(\frac{3\pi t}{5}\right) = 51.5$

$$\Rightarrow \sin\left(\frac{3\pi t}{5}\right) = \frac{51.5-56}{9} = \frac{-4.5}{9} = -\frac{1}{2}$$

$$\Rightarrow \frac{3\pi t}{5} = \frac{7\pi}{6} \Rightarrow t = \frac{7\pi}{6}\cdot\frac{5}{3\pi} = \frac{35}{18} = 1\frac{17}{18}\ months$$

d) $s(12) = 56 + 9\sin\left(\frac{3\pi(12)}{5}\right) = 56 + 9\sin\left(\frac{-4\pi}{5}\right)$

$$= 56 + 9\sin\left(\frac{-4\pi}{5}\right) = 56 - 9(0.5878) = 50.7098$$

After one year the sale reaches 50,709 birr.

I. Sum and Difference formulas

- $\sin(A + B) = \sin A \cos B + \cos A \sin B.$
- $\sin(A - B) = \sin A \cos B - \cos A \sin B.$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B.$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B.$
- $\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$
- $\tan(A - B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$

II. Double angle formula

- $\sin 2A = \sin(A + A) = 2\sin A \cos A$
- $\cos 2A = \cos(A + A) = \cos^2 A - \sin^2 A$
- $\tan 2A = \tan(A + A) = \dfrac{2\tan A}{1 - \tan^2 A}$

III. Half angle formula

- $\sin^2\left(\frac{A}{2}\right) = \dfrac{1-\cos A}{2} \Rightarrow \sin\left(\frac{A}{2}\right) = \pm\sqrt{\dfrac{1-\cos A}{2}}$
- $\cos^2\left(\frac{A}{2}\right) = \dfrac{1+\cos A}{2} \Rightarrow \cos\left(\frac{A}{2}\right) = \pm\sqrt{\dfrac{1+\cos A}{2}}$

- $\tan^2\left(\frac{A}{2}\right)=\frac{1-\cos A}{1+\cos A}\Rightarrow \tan\left(\frac{A}{2}\right)=\pm\sqrt{\frac{1-\cos A}{1+\cos A}}=\frac{\sin A}{1+\cos A}$

Find the exact value:

a) $\sin 75^0$ d) $\sin 105^0$ g) $\cos\frac{\pi}{12}$ j) $\sec\frac{11}{12}\pi$

b) $\sin 15^0$ e) $\tan(15^0)$ h) $\tan\frac{17\pi}{12}$

c) $\cos 165^0$ f) $\tan 195^0$ i) $\cot\frac{19\pi}{12}$

Solution:

a) $\sin 75^0=\sin(45^0+30^0)=\sin 45^0\cos 30^0+\cos 45^0\sin 30^0$

$$=\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2}+\frac{\sqrt{2}}{2}\cdot\frac{1}{2}=\frac{\sqrt{6}+\sqrt{2}}{4}$$

b) $\sin 15^0=\sin(45^0-30^0)=\sin 45\cos 30^0-\cos 45^0\sin 30^0$

$$=\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2}+\frac{\sqrt{2}}{2}\cdot\frac{1}{2}=\frac{\sqrt{6}-\sqrt{2}}{4}$$

c) $\cos 165^0=\cos(180^0-150^0)=\cos 180^0\cos 15^0+\sin 180^0\sin 15^0$

$$=-\cos 15^0$$

$$\Rightarrow -\cos 15^0=-\cos(45^0-30^0)$$

$$=-(\cos 45\cos 30+\sin 45^0\sin 30^0)$$

$$=-\left(\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2}+\frac{\sqrt{2}}{2}\cdot\frac{1}{2}\right)=\left(-\frac{\sqrt{6}+\sqrt{2}}{4}\right)$$

$$\Rightarrow \cos 165^0=\cos(180^0-15^0)=-\cos 15^0=\frac{-\sqrt{6}-\sqrt{2}}{4}$$

d) $\sin 105^0=\sin(60^0+45^0)=\sin 60^0\cos 45^0+\cos 60^0\sin 45^0=\frac{\sqrt{6}+\sqrt{2}}{4}$

Note: $\cos 105^0=\frac{\sqrt{2}-\sqrt{6}}{4}$

e) $\tan 150^0 = \tan(180^0 - 30^0) = \dfrac{\tan 180^0 - \tan 30^0}{1 + \tan 180^0 - \tan 30}$

f) $\tan 195^0 = \tan(150^0 + 45^0)$

$$= \frac{\tan 150^0 + \tan 45^0}{1 - \tan 150^0 \cdot \tan 45^0} = \frac{-\frac{\sqrt{3}}{3} + 1}{1 + \frac{\sqrt{3}}{3}} = 2 - \sqrt{3}$$

g) $\cos\left(\frac{\pi}{12}\right) = \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \cos\frac{\pi}{3}\cos\frac{\pi}{4} + \sin\frac{\pi}{3}\sin\frac{\pi}{4}$

$$= \frac{1}{2}, \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}$$

h) $\tan\left(\frac{17\pi}{12}\right) = \tan\left(\frac{7\pi}{12} - 2\pi\right) = \tan\left(\frac{-7\pi}{12}\right) = -\tan\left(\frac{\pi}{3} + \frac{\pi}{4}\right)$

$$\Rightarrow \frac{-\left(\tan\frac{\pi}{3} + \tan\frac{\pi}{4}\right)}{1 - \tan\frac{\pi}{3}\tan\frac{\pi}{4}} = -\left(\frac{\sqrt{3}+1}{1-\sqrt{3}}\right) = -\sqrt{3} - 2$$

i) $\cot\left(\frac{19}{12}\pi\right) = \cot\left(\frac{19\pi}{12} - \pi\right) = \cos\left(\frac{7\pi}{12}\right) = \frac{1}{\tan\left(\frac{7\pi}{12}\right)} = 2 - \sqrt{3}$

j) $\sec\left(\frac{11\pi}{12}\right) = -\sec\left(\pi - \frac{11\pi}{12}\right) = -\sec\frac{\pi}{12} = -\sec\left(\frac{\pi}{3} - \frac{\pi}{4}\right)$

$$= \frac{-1}{\cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right)} = \frac{-1}{\cos\frac{\pi}{3}\cos\frac{\pi}{4} + \sin\frac{\pi}{3}\sin\frac{\pi}{4}} = \frac{-4}{\sqrt{6} + \sqrt{2}} = \sqrt{2} - \sqrt{6}$$

Solved problem

63. Simplify each of the following expressions

a) $\sin(x + \pi)$ d) $\csc\left(x + \frac{\pi}{2}\right)$ g) $\cos\left(x - \frac{\pi}{2}\right)$

b) $\sin\left(x+\frac{\pi}{2}\right)$ e) $\cos\left(x+\frac{\pi}{2}\right)$ h) $\cos(x + k\pi)$, $k\in Z$

c) $\tan\left(x+\frac{\pi}{2}\right)$ f) $\sec\left(x-\frac{\pi}{2}\right)$ i) $\sin(x + k\pi)$

Solution:

a) $\sin(x+\pi) = \sin x \cos\pi + \cos x \sin\pi = -\sin x$

b) $\sin\left(x+\frac{\pi}{2}\right) = \sin x \cos\frac{\pi}{2} + \cos x \sin\frac{\pi}{2} = 0 + \cos x = \cos x$

c) $\tan\left(x+\frac{\pi}{2}\right) = \frac{\sin\left(x+\frac{\pi}{2}\right)}{\cos\left(x+\frac{\pi}{2}\right)} = \frac{\cos x}{\cos x\cos\frac{\pi}{2} - \sin x\sin\frac{\pi}{2}} = \frac{\cos x}{-\sin x} = -\cot x$

d) $\csc\left(x+\frac{\pi}{2}\right) = \frac{1}{\sin\left(x+\frac{\pi}{2}\right)} = \frac{1}{\cos x} = \sec x$

e) $\cos\left(x+\frac{\pi}{2}\right) = \cos x \cos\frac{\pi}{2} - \sin x \sin\frac{\pi}{2} = 0 - \sin x = -\sin x.$

f) $\sec\left(x-\frac{\pi}{2}\right) = \frac{1}{\cos\left(x-\frac{\pi}{2}\right)} = \frac{1}{\sin x} = \csc x$

g) $\cos\left(x-\frac{\pi}{2}\right) = \cos x \cos\frac{\pi}{2} + \sin x \sin\frac{\pi}{2} = \sin x.$

h) $\cos(x + k\pi) = \cos x \cos\pi k - \sin x \sin\pi k = (-1)^k \cos x$, k is integer

i) $\sin(x + k\pi) = \sin x \cos\pi k + \cos x \sin\pi k = (-1)^k \sin x.$

63. Simplify

a) $\frac{\cos 48^0 \sin 12^0 + \sin 48^0 \cos 12^0}{\cos 72^0 \cos 12^0 + \sin 72^0 \sin 12^0}$

b) $\frac{\tan 189^0 - \tan 144^0}{1 + 189^0 \tan 144^0}$

c) $\cos 130^0 \cos 50^0 - \sin 130^0 \sin 50^0$

Solution:

a) $\Rightarrow \dfrac{\sin(48^0+12^0)}{\cos(72-12)} = \dfrac{\sin 60^0}{\cos 60} = \tan 60^0 = \sqrt{3}$

b) $\tan(189^0 - 144^0) = \tan 45^0 = 1$

c) $\cos(130^0 + 50^0) = \cos 180^0 = -1$

Sum – to – product formula

- $\sin x + \sin y = 2\sin\left(\dfrac{x+y}{2}\right)\cos\left(\dfrac{x-y}{2}\right)$
- $\sin x - \sin y = 2\cos\left(\dfrac{x+y}{2}\right)\sin\left(\dfrac{x-y}{2}\right)$
- $\cos x + \cos y = 2\cos\left(\dfrac{x+y}{2}\right)\cos\left(\dfrac{x-y}{2}\right)$
- $\cos x - \cos y = -2\sin\left(\dfrac{x+y}{2}\right)\sin\left(\dfrac{x-y}{2}\right)$

64. Simplify: a) $\dfrac{\sin 2x + \sin 4x}{\cos 2x - \cos 4x} = \dfrac{2\sin\left(\dfrac{4x+2x}{2}\right)\cos\left(\dfrac{2x-4x}{2}\right)}{-2\sin\left(\dfrac{2x+4x}{2}\right)\sin\left(\dfrac{2x-4x}{2}\right)}$

$\Rightarrow \dfrac{2\sin 3x\cos x}{-2\sin 3x\sin(-x)} = \dfrac{-\cos x}{\sin(-x)} = \dfrac{\cos x}{\sin x} = \cot x.$

b) $\cos 195^0 + \cos 105^0 = 2\cos$

$\left(\dfrac{195^0+105^0}{2}\right)\cos\left(\dfrac{195^0-105^0}{2}\right) = 2\cos 150^0\cos 45^0$

$\Rightarrow 2\left(-\dfrac{\sqrt{3}}{2}\right)\left(\dfrac{\sqrt{2}}{2}\right) = \dfrac{-\sqrt{6}}{2}$

64) Find the exact value

a) $\tan^{-1}\left(-\sqrt{3}\right)$

b) $\sec^{-1}(-2)$

c) $\sin\left(\cos^{-1}\left(-\dfrac{1}{2}\right)\right)$

d) $\sin^{-1}\left(\sin\dfrac{5\pi}{4}\right)$

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Solution:

a) $\tan^{-1}\left(-\sqrt{3}\right) = -\tan^{-1}\left(\sqrt{3}\right) = \frac{-\pi}{3}$

b) $\sec^{-1}(-2) = \pi - \sec^{-1}(2) = \pi - \cos^{-1}\left(\frac{1}{2}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$

c) $\sin\left(\cos^{-1}\left(-\frac{1}{2}\right)\right) = \sin$

$\left(\pi - \cos^{-1}\left(\frac{1}{2}\right)\right) = \sin\left(\pi - \frac{\pi}{3}\right) = \sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$

d) $\sin^{-1}\left(\sin\frac{5\pi}{4}\right) = \sin^{-1}$

$\left(-\sin\left(\frac{5\pi}{4} - \pi\right)\right) = \sin^{-1}\left(\frac{-\sqrt{2}}{2}\right) = \frac{-\pi}{4}$

65) If $\sin\theta - \cos\theta = ½$ then what is the value of $\sin 2\theta$ . . . UEE

A. $\frac{3}{8}$ B. $\frac{3}{4}$ C. $\frac{5}{8}$ D. $\frac{5}{4}$

Solution:

$\Rightarrow (\sin\theta - \cos\theta)^2 = ¼ \Rightarrow \sin^2\theta - 2\sin\theta\cos\theta + \cos^2\theta = ¼$

$\Rightarrow 1 - 2\sin\theta\cos\theta = ¼ \Rightarrow 2\sin\theta\cos\theta = 3/8$

$\Rightarrow 2\sin\theta\cos\theta = \sin 2\theta = 3/8$

Answer: A

66) $\cos(\sin^{-1}(x))$ is equal to …UEE

A. $\sqrt{x^2+1}$ C. $\frac{1}{1-x^2}$

B. $\sqrt{1-x^2}$ D. $\frac{1}{\sqrt{x^2+1}}$

Solution: Let $\sin^{-1} x = \theta \Rightarrow \sin\theta = x$, $\cos\theta = \sqrt{1-\sin^2\theta}$

$\Rightarrow \cos(\sin^{-1} x) = \cos\theta = \sqrt{1-x^2}$,

Answer: B

67) If $\sin 37 = 0.6$ and $\sin 53^0 = 0.8$, then $\cos 16^0$ is ... UEE
A. 0.02 B. 0.90 C. 0.48 D. 0.96

Solution: $\cos 16^0 = \cos(53-37) = \cos(53)\cos(37) + \sin(53)\sin(37)$
$\Rightarrow \cos 16^0 = \sin 37 \sin 53 + \sin 53 \sin 37$
$= 2(\sin 37 \sin 53) = 2(0.6)(0.8) = 0.96$

68) If $\cos\theta = \frac{-5}{13}$ *for* $\frac{\pi}{2} < \theta < \pi$, then $\sin 2\theta$ is equal to... EHEECE

A. $\frac{120}{169}$ B. $\frac{-120}{240}$ C. $\frac{169}{240}$ D. $\frac{10}{13}$

Solution:

$$\text{If } \cos\theta = \frac{-5}{13} \Rightarrow \sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\left(\frac{25}{169}\right.} = \frac{12}{13}$$

$$\Rightarrow \sin 2\theta = \sin(\theta+\theta) = 2\sin\theta\cos\theta = 2\left(\frac{12}{13}\right)\left(\frac{-5}{13}\right) = \frac{-120}{169}$$

Answer:

69) Given $\Delta ABC\ \overline{AB} = 30\ unit$, $\overline{AC} = 22$ units and $\overline{BC} = 18$ units, then cos B is equal to :

A. $\frac{37}{54}$ B. $\frac{54}{37}$ C. $\frac{3}{5}$ D. $\frac{11}{15}$

Solution: Let, $\overline{BC} = a = 18$, $\overline{AC} = b = 22$, and $\overline{AB} = c = 30$

$$\Rightarrow \cos B = \frac{a^2+c^2-b^2}{2ac} = \frac{18^2+30^2-22^2}{2(18)(30)} = \frac{324+900-484}{1080} =$$

70) In ΔABC, $m(<A) = 30^0$, $m(<B) = 45^0$ and BC = 12 units
Then how long is $\overline{AC}$...

A. $12\sqrt{3}$ unit B. $4\sqrt{6}$ C. $6\sqrt{2}$ unit D. $12\sqrt{2}$ unit

Solution:

$$\frac{a}{\sin A} = \frac{b}{\sin B} \Rightarrow \frac{12}{\sin 30^0} = \frac{AC}{\sin 45^0} \Rightarrow AC = \frac{12}{\frac{1}{2}} \cdot \frac{\sqrt{2}}{2} = 12\sqrt{2}$$

71) If $\theta = \arccos\left(\frac{1}{2}\right)$, what is the value of $\tan\theta$ . . . UEE 2002.

A. $\frac{1}{\sqrt{3}}$ B. $\sqrt{3}$ C. $\frac{\sqrt{3}}{2}$ D. 1

Solution: $\theta = \cos^{-1}\left(\frac{1}{2}\right) \Rightarrow \cos\theta = \frac{1}{2}$,

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{1}{4}}$

$\Rightarrow \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$,

Answer: B

72) If $x = \arcsin\left(\frac{3}{4}\right)$, the value of $\cos(2x)$. . . UEE

A. $\frac{3}{4}$ B. $\frac{-3}{4}$ C. $\frac{-1}{8}$ D. $\frac{1}{8}$

Solution: $x = \sin^{-1}\left(\frac{3}{4}\right) \Rightarrow \sin x = \frac{3}{4} \Rightarrow \cos x = \sqrt{1-\sin^2 x}$

$= \sqrt{1-\frac{9}{16}} = \frac{\sqrt{7}}{4}$

$\Rightarrow \cos 2x = \cos(x+x) = \cos^2 x - \sin^2 x = \frac{7}{16} - \frac{9}{10} = \frac{-2}{16} = \frac{-1}{8}$

73) $\text{Sec}\left(\tan^{-1}\left(-\sqrt{3}\right)\right)$ is equal to:

Answer: C

A. $\frac{-3\sqrt{3}}{2}$ B. -2 C. $\frac{3\sqrt{3}}{2}$ D. 2

Solution: $\sec\left(\tan^{-1}\left(-\sqrt{3}\right)\right) = \sec\left(\frac{-\pi}{3}\right) = 2$.

74) The Solution set of the equation: $2\cos^2 x - \sin^2$

Answer: D

A. $\left\{\frac{\pi}{2}+k\pi\right\}$ C. $\left\{\frac{\pi}{2}+k\pi\right\}$

B. $\left\{\frac{\pi}{4}+k\pi,\frac{\pi}{2}+k\pi\right\}$ D. $\left\{\frac{\pi}{4}+k\pi,\frac{\pi}{2}+2k\pi\right\}$

Solution: $2\cos^2 x - \sin 2x = 0 \Leftrightarrow 2\cos^2 x - 2\sin x \cos x = 0$

$\Rightarrow 2\cos x(\cos x - \sin x) = 0 \Rightarrow 2\cos x = 0$ or $\cos x = \sin x$

$\Rightarrow x = \frac{\pi}{2}+2\pi k, x = \frac{\pi}{4}+k\pi$

Answer: D

75) If $\theta = \arcsin\left(\frac{1}{2}\right)$, then $\tan(2\theta)$ is equal to … UEE 2003

A. $\frac{1}{\sqrt{3}}$ B. 1 C. $\sqrt{3}$ D. $\frac{\sqrt{3}}{2}$

Solution: $\sin^{-1}\left(\frac{1}{2}\right) = \theta \Rightarrow \sin\theta = \frac{1}{2}$,

$\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-\frac{1}{4}} = \frac{\sqrt{3}}{2}$

$$\Rightarrow \tan 2\theta = \frac{\sin 2\theta}{\cos 2\theta} = \frac{2\sin\theta\cos\theta}{\cos^2\theta - \sin^2\theta} = \frac{2\left(\frac{1}{2}\right)\frac{\sqrt{3}}{2}}{\frac{3}{4}-\frac{1}{4}} = \frac{\sqrt{3}}{2}\cdot(2) = \sqrt{3}$$

76) What is the amplitude and period respectively of the graph $f(x) = 2 - 3\sin\left(\frac{x+\pi}{4}\right)$…

A. $2, \frac{\pi}{4}$ B. -3, 8π C. 3, $\frac{\pi}{2}$ D. 3, 8

Solution: Amplitude, $a = |-3| = 3$, period, $p = \frac{2\pi}{k} = \frac{2\pi}{\frac{1}{4}} = 8$

Answ

77) Which of the following is the Solution set of $\sqrt{3}\sin 2x = \cos 2x$ on $[0,\pi]$...

A. $\left\{\frac{\pi}{6},\frac{\pi}{12}\right\}$ B. $\left\{\frac{\pi}{6},\frac{\pi}{3}\right\}$ C. $\left\{\frac{\pi}{12},\frac{7\pi}{12}\right\}$ D. $\left\{\frac{\pi}{6},\frac{2}{3}\pi\right\}$

Solution: $\sqrt{3}\sin 2x = \cos 2x \Rightarrow \tan 2x = \frac{\sin 2x}{\cos 2x} = \frac{1}{\sqrt{3}}$

$\Rightarrow 2x = \frac{\pi}{6},\frac{7\pi}{6} \Rightarrow x = \left\{\frac{\pi}{12},\frac{7\pi}{12}\right\}$ **Answer: C**

78) A 1 meter high pole is fixed vertically on the ground at a point $50\sqrt{3}$ m far from the foot of building. If the angle of elevation from the top of the pole to the top of building is 30^0. What is the height of the building... UEE 2003.

A. 50m B. 51m C. 150m D. 151m

Solution: $\tan 30^0 = \frac{d}{50\sqrt{3}} \Rightarrow d = 50\sqrt{3}.\frac{1}{\sqrt{3}} = 50m$

$\Rightarrow h = d + y = 50m + 1m = 51m$

Answer: B

79) Let $f(x) = -3\sin\left(\frac{1}{2}x\right)$. which of the following is true...UEE2002

A. The amplitude is –3 C. f is decreasing on $[-2\pi,0]$

B. The range is [-1, 1] D. f is increasing on $[\pi, 3\pi]$

Solution: Amplitude, $a = |-3| = 3$, Range = [-3,3]

- f is increasing on ½ $x = \left[\frac{\pi}{2},\frac{3\pi}{2}\right] \Rightarrow [\pi,3\pi]$

Answer: D

80) In the interval $[0,2\pi]$, what is the Solution set of the equation $\cos^2 x + \sin x \cos x = 1$ ...UEE

Solution: $\cos^2 x + \sin x \cos x = 1 \Rightarrow \sin x\cos x = 1 - \cos^2 x = \sin^2 x$

$\Rightarrow \sin x \cos x = \sin^2 x \Rightarrow \sin^2 x - \sin x \cos x = 0 \Rightarrow \sin x(\sin x - \cos x) = 0$

$\Rightarrow \sin x = 0, \sin x = \cos x \Rightarrow x = \left\{0,\frac{\pi}{4},\frac{5\pi}{4},\pi\right\}$ **Answer: A**

81) Given that $\tan\theta = \frac{-4}{3}$, θ is in quadrant II, and $\tan\beta = \frac{-5}{12}$ for β in quadrant IV then find

A. $\sin(\theta+\beta)$ B. $\cos(\theta+\beta)$ C. quadrant of $(\theta-\beta)$

Solution:

$$\tan\theta = \frac{-4}{3} \Rightarrow \sin\theta = \frac{4}{5} \text{ and } \cos\theta = \frac{-3}{5} \quad \tan\beta = \frac{-5}{12} \Rightarrow$$

$$\sin\beta = \frac{-5}{13} \text{ and } \cos\beta = \frac{12}{13}$$

$$\Rightarrow \sin(\theta+\beta) = \sin\theta\cos\beta + \cos\theta\sin\beta = \frac{4}{5}\cdot\frac{12}{13} + \frac{-3}{5}\cdot\frac{-5}{13} = \frac{63}{65}$$

$$\Rightarrow \cos(\theta+\beta) = \cos\theta\cos\beta - \sin\theta\sin\beta = \left(\frac{-3}{5}\right)\left(\frac{12}{13}\right) - \left(\frac{4}{5}\right)\left(\frac{-5}{13}\right) = \frac{-16}{65}$$

- $\sin(\theta+\beta) > 0$ and $\cos(\theta+\beta) < 0 \Rightarrow (\theta+\beta)$ lies on II quadrant

82) Exact value of $\cos\left(\sin^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{-4}{5}\right)\right)$

A. -1 B. 1 C. ½ D. $-½$

Solution:

$$\text{Let } \sin^{-1}\left(\frac{3}{5}\right) = \theta \Rightarrow \sin\theta = \frac{3}{5},\ \cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-\frac{-9}{25}} = \frac{4}{5}$$

$$\text{Let } \cos^{-1}\left(\frac{-4}{5}\right) = \beta \Rightarrow \cos\beta = \frac{-4}{5}, \sin\beta = \sqrt{1-\cos^2\beta} = \sqrt{1-\frac{16}{25}} = \frac{3}{5}$$

$$\Rightarrow \cos\left(\sin^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{-4}{5}\right)\right) \Rightarrow \cos(\theta+\beta) = \cos\theta\cos\beta - \sin\theta\sin\beta$$

$$\Rightarrow \left(\frac{4}{5}\right)\left(\frac{-4}{5}\right) - \left(\frac{3}{5}\right)\left(\frac{3}{5}\right) = \frac{-16-9}{25} = -1$$

Answer: A

83) If $\cos\theta = \frac{-4}{5}$ *for*, $\pi < \beta < \frac{3\pi}{2}$ *then find*

A. $\sin\left(\frac{\theta}{2}\right)$ B. $\cos\left(\frac{\theta}{2}\right)$ C. $\tan\left(\frac{\theta}{2}\right)$ D. $\cos2\theta$ E. $\sin2\theta$

Solution: $\pi < \theta < \frac{3\pi}{2} \Rightarrow \frac{\pi}{2} < \frac{\theta}{2} < \frac{3\pi}{4}$

$\Rightarrow \sin\left(\frac{\theta}{2}\right) > 0$ and $\cos\left(\frac{\theta}{2}\right) < 0$

a) $\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = -\sqrt{\frac{1+\frac{4}{5}}{2}} = \frac{3}{\sqrt{10}}$

b) $\cos\left(\frac{\theta}{2}\right) = -\sqrt{\frac{1+\cos\theta}{2}} = \sqrt{\frac{1-\frac{4}{5}}{2}} = -\sqrt{\frac{3}{10}} = \frac{-1}{\sqrt{10}}$

c) $\tan\left(\frac{\theta}{2}\right) = -\frac{\sin\left(\frac{\theta}{2}\right)}{\cos\left(\frac{\theta}{2}\right)} = \frac{\frac{3}{\sqrt{10}}}{\frac{-1}{\sqrt{10}}} = -3$

d) $\cos 2\theta = \cos^2\theta - \sin^2 = \left(\frac{-4}{5}\right)^2 - \left(\frac{-3}{25}\right)^2 = \frac{16-9}{25} = \frac{7}{25}$

e) $\sin 2\theta = \sin(\theta+\theta) = 2\sin\theta\cos\theta = 2\left(\frac{-3}{5}\right)\left(\frac{-4}{25}\right) = \frac{24}{25}$

84) Let $\sin\left(2x - \frac{\pi}{4}\right) = d$ *then solve for*

a) $d = 0$ b) $d = 1$ c) $d = ½$

Solution: $\sin\left(2x - \frac{\pi}{4}\right) = 0 \Rightarrow 2x - \frac{\pi}{4} = 0, \pi \ldots \text{period} = \pi$

$\Rightarrow 2x = \frac{\pi}{4}, \frac{5\pi}{4} \ldots, x = \frac{\pi}{8} + \pi k, \frac{5\pi}{4} + \pi k$

$\therefore SS = \left\{\frac{\pi}{8} + \pi k; \frac{5\pi}{4} + \pi k\right\}$

b) $\sin\left(2x-\frac{\pi}{4}\right)=1 \Rightarrow 2x-\frac{\pi}{4}=\frac{\pi}{2} \Rightarrow 2x=\frac{3\pi}{4}+2\pi k$

$\Rightarrow x=\frac{3\pi}{8}+\pi k$

d) $\sin\left(2x-\frac{\pi}{4}\right)=\frac{1}{2} \Rightarrow 2x-\frac{\pi}{4}=\frac{\pi}{6},\frac{5\pi}{6}$

$\Rightarrow 2x=\frac{\pi}{6}+\frac{\pi}{4}+2\pi k,\frac{5\pi}{6}+\frac{\pi}{4}+2\pi k$

$\Rightarrow x=\frac{10\pi}{48}+\pi k,\frac{15\pi}{48}+\pi k=\left\{\frac{5\pi}{24}+\pi k,\frac{5\pi}{16}+\pi k\right\}$

Supplementary exercise

1. Find all Solutions of the equation in the interval, $[0,2\pi]$

a) $\sin\left(x+\frac{\pi}{3}\right)+\sin\left(x-\frac{\pi}{3}\right)=1$

b) $\cos\left(x+\frac{\pi}{4}\right)-\cos\left(x-\frac{\pi}{4}\right)=1$

c) $\tan(x+\pi)+2\sin(x+\pi)=0$

2. Find the amplitude, phase shift, period for the following functi

a) $y=2\sin\left(x-\frac{\pi}{2}\right)$

b) $y=\cos\left(2x-\frac{\pi}{4}\right)$

c) $y=-4\sin\left(\frac{2x}{3}+\frac{\pi}{6}\right)$

d) $y=\frac{5}{4}\cos(3x-2\pi)$

3. Find phase shift and period for the function

a) $y=2\tan\left(2x-\frac{\pi}{4}\right)$

b) $y=-3\csc\left(\frac{x}{3}+\pi\right)$

4. Graph one full period of each function.

a) $y=\sin\left(x-\frac{\pi}{2}\right)$

b) $y=\cos\left(\frac{x}{2}+\frac{\pi}{3}\right)$

d) $y=-\sin(\pi x+1)-2$

e) $y=4\cos(\pi x-2)+1$

c) $y = \tan\left(x - \frac{\pi}{4}\right)$

5. Find the equation of

a) the cosine function with amplitude 3 period 3π, and phase shift $\frac{-\pi}{4}$

b) the tangent function with period, 2π, phase shift $\frac{\pi}{2}\ \frac{\pi}{2}$

c) the secant function with period 4π and phase shift $\frac{3\pi}{4}$

Find the exact value

a) $\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)$

b) $\tan^{-1}(-1)$

c) $\sec^{-1}(2)$

d) $\cos(\sec^{-1}2)$

e) $\cos(2\sin^{-1}\left(\frac{\sqrt{2}}{2}\right))$

f) $\sin(2\sin^{-1}\left(\frac{4}{2}\right))$

g) $\sin^{-1}[\cos\left(\frac{-2\pi}{3}\right)]$

h) $\sin^{-1}(\frac{1}{2}\sin^{-1}x)$

Find an algebraic expression for each of the following

a) $\csc(\cos^{-1}(x))$

b) $\sin(2\cos^{-1}x)$

c) $\sin(\cos^{-1}x - \sin^{-1}3x)$

Find the exact value of the given expression.

a) $\sin\left(\sin^{-1}\left(\frac{2}{3}\right) + \cos^{-1}\left(\frac{1}{2}\right)\right)$ b) $\tan\left(\cos^{-1}\left(\frac{1}{2}\right) - \sin^{-1}\left(\frac{3}{4}\right)\right)$

Solve the equation for x algebraically.

a) $\tan^{-1}x = \sin^{-1}\left(\frac{24}{25}\right)$

b) $\cos^{-1}(x - \frac{1}{2}) = \frac{\pi}{3}$

c) $\sin^{-1}x + \cos^{-1}\left(\frac{4}{5}\right) = \frac{\pi}{6}$

d) $\cos^{-1}x + \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{2}$

10. Solve for y interms of x

a) $5x = \tan^{-1}3y$ b) $x - \frac{\pi}{3} = \cos^{-1}(y-3)$

11. solve the equation

a) $y = \sec\left(\sin^{-1}\left(\frac{12}{13}\right)\right)$ d) $y = \cos\left[\sin^{-1}\left(-\frac{3}{5}\right) + \cos^{-1}\left(\frac{5}{13}\right)\right]$

b) $2\sin^{-1}(x-1) = \frac{\pi}{3}$ e) $y = \cos\left(\sin^{-1}\left(\frac{3}{5}\right)\right)$

c) $\sin^{-1}x + \cos^{-1}\frac{4}{5} = \frac{\pi}{2}$

12. Given that $\tan A = \frac{12}{5}$ and A is in quadrant II and that $\cos B = \frac{9}{4}$, and B is in quadrant I.

Find the following

a) $\cos(A-B)$ C) $\sin(A-B)$

b) $\tan(A+B)$ D) quadrant of $A-B$

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